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J Pediatr Adolesc Gynecol. Author manuscript; available in PMC 2024 June 24.

Published in final edited form as:
J Pediatr Adolesc Gynecol. 2024 June ; 37(3): 315–322. doi:10.1016/j.jpag.2024.02.004.

Ovarian Morphology in Girls Longitudinal Cohort Study: Pilot Evaluation of Ovarian Morphology as a Biomarker of Reproductive and Metabolic Features during the First Gynecological Year

Heidi Vanden Brink, PhD, MS[1,2,3,*], **Tania S. Burgert, MD, ScD**[2], **Romina Barral, MD, MSCR**[4,5,6], **Anushka Malik, BS**[1], **Manasa Gadiraju**[5], **Marla E. Lujan, PhD, MS**[1,**]

[1]Division of Nutritional Sciences, Cornell University, Ithaca, New York

[2]Division of Endocrinology, Children's Mercy Kansas City, Kansas City, Missouri

[3]Department of Nutrition, Texas A&M University, College Station, Texas

[4]Division of Adolescent Medicine, Children's Mercy Kansas City, Kansas City, Missouri

[5]University of Missouri Kansas City School of Medicine, Kansas City, Missouri

[6]University of Kansas Medical Center, Kansas City, Kansas

Abstract

Objective: The objective was to establish whether aspects of ovarian morphology correlate with reproductive and metabolic features during the first postmenarcheal year using data from the Ovarian Morphology in Girls (OMG!) cohort study. The feasibility of transabdominal ultrasonography to assess ovarian features was also determined.

Methods: Healthy adolescent females enrolled in a prospective cohort study. Study visits occurred at 6–10, 11–13, 17–19, and 23–25 months postmenarche and entailed a physical exam,

[*]*Address correspondence to*: Heidi Vanden Brink, PhD, MS, Assistant Professor, Department of Nutrition, 373 Olsen Blvd, Texas A&M University, College Station, TX, 77845. heidi.vandenbrink@ag.tamu.edu (Heidi Vanden Brink). [**]*Address correspondence to*: Marla E. Lujan, PhD, MS, Associate Professor, Division of Nutritional Sciences, Savage Hall, Cornell University, Ithaca, NY, 14850. marla.lujan@cornell.edu (Marla E. Lujan).

Conflicts of Interest
HVB has received funding from the Canadian Institutes of Health Research (FRN 146182), a grant from the Cornell University Office of Academic Integration, and a CMRIF Pilot Award from Cornell University. MEL has received R-01 and R-21 funding from the National Institutes of Health, a grant from the Cornell University Office of Academic Integration, a CMRIF Pilot Award from Cornell University, and payment or honoraria from the Society of Obstetrics and Gynecology of Canada; serves in a leadership role in the Androgen Excess and PCOS Society; and has received equipment, materials, gifts, or other services from the PCOS Awareness Association. TSB has received R-21 funding from the National Institutes of Health. RB has received R-21 and K-23 funding from the National Institutes of Health. AM and MG have no disclosures.

Attestation Statement
The subjects in this study have not concomitantly been involved in other trials. Data regarding any of the subjects in the study have not been previously published unless specified.

Implications and Contribution
We describe the establishment of a longitudinal domestic cohort to characterize ovarian morphology in the early postmenarcheal years. We demonstrate that known associations between ovarian morphology with reproductive and metabolic features are apparent within the first year of gynecological life.

Supplementary materials
Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jpag.2024.02.004.

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transabdominal ultrasound, and fasting blood draw. Participants maintained menstrual diaries throughout the study. The present analysis reflects participants who completed the study visit at 6–10 months postmenarche. Associations between ovarian morphology or average cycle length with reproductive and metabolic features were assessed by Spearman correlations and linear regression.

Results: Forty participants enrolled in the OMG! study. Thirty-one participants initiated study procedures at 6–10 months postmenarche, and data were available for analysis for 29 participants. Image quality was judged as partially visible or excellent in 90% of the left and 78% of the right ovaries assessed, with all images collected having sufficient image quality to provide measurements of at least 1 ovarian marker. The follicle number per ovary and ovarian volume were positively associated with anti-Müllerian hormone levels and negatively associated with fasting insulin. The average cycle length was only associated negatively with triglycerides.

Conclusion: Transabdominal ultrasonography in the early postmenarcheal period provides sufficient resolution to enable estimations of antral follicle count and ovarian size. Ovarian features in early gynecological life may correspond with measures of reproductive and metabolic function.

Keywords

Menarche; Ovary; Menstrual cycle; Adolescence; Ultrasound

Introduction

The early postmenarcheal years represent a critical window of reproductive maturation. The trajectory of menstrual cyclicity and associated reproductive hormone milieu have been characterized[1–3] and support that menstrual irregularity in adolescence may be associated with continued cycle irregularity, reduced fertility, and polycystic ovary syndrome (PCOS) into adulthood.[4,5] However, menstrual cyclicity can be unpredictable in the first gynecological years, owing to the immaturity of the reproductive axis, and does not serve as a useful biomarker of early reproductive dysfunction.[6] Ultimately, this lack of early biomarkers results in delayed diagnosis, intervention, and possible prevention of reproductive disturbances such as PCOS, which are known to manifest during adolescence.[7]

Ovarian morphology on ultrasonography could potentially serve as a noninvasive biomarker of early reproductive dysfunction. A variety of ovarian features, including antral follicle number per ovary (FNPO), ovarian volume (OV), stromal area, and echogenicity, have predictive power for hyperandrogenic causes of anovulation in adults,[8–10] and preliminary evidence supports a role of ovarian enlargement in the identification of PCOS in adolescents.[11–13] The number of antral follicles and ovarian size reflect the degree of reproductive symptomology in both adult[14] and adolescent[11] PCOS, and accumulating evidence supports their ability to reflect nutritional[16] and metabolic status.[15,16] Studies reporting the manifestation of reproductive dysfunction in early gynecological life have focused on changes beginning 1 or more years after the onset of menarche,[17,18] leaving the first menarcheal year largely unexplored. These studies are also dated, did not have the benefit of modern ultrasound technology, and were conducted in non-domestic populations that may not reflect the current demographic makeup of the United States.

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To that end, we have established a longitudinal cohort of US-based adolescents as part of the Ovarian Morphology in Girls (OMG!) study to resolve the trajectory of ovarian morphology alongside menstrual cyclicity in early gynecological life. Herein, we assessed the degree to which aspects of ovarian morphology correlate with reproductive and metabolic features during the first postmenarcheal year. The feasibility of transabdominal ultrasonography to assess ovarian features during this early postmenarcheal period was also determined.

Methods

Prospective Cohort Study Design

Adolescent females were enrolled in a prospective cohort study (ie, OMG!) designed to monitor longitudinal changes in ovarian morphology alongside reproductive and metabolic parameters during the first 2 years postmenarche. The a priori recruitment goal was 22 adolescents self-reporting regular menstrual cycles on enrollment and 26 adolescents reporting irregular cycles, with body mass index balanced across menstrual cycle status groups. Recruitment goals assumed that 10% of adolescents with irregular cycles at entry would transition to regular cycles by 2 years postmenarche,[3,19] with an attrition rate of 15%.[20] A sample size of 21 adolescents per outcome group (menstrual cycle status at 2 years postmenarche) would enable detection of an effect size of 0.9 in OV, or a difference of 1.7–2.6 cm^3, between groups at an alpha level of 0.05 with 80% power given previously published data in non-US cohorts.[21,22]

Recruitment began in January 2020 and concluded in February 2021 at Children's Mercy Kansas City (CMKC, Kansas City, MO) and in May 2021 at Cornell University (Ithaca, NY). Participants were recruited using advertisements disseminated through community and professional listservs and social media applications in the Tompkins (NY), Jefferson (KS), Jackson (MO), Clay (MO), Platte (MO), and Wyandotte (KS) counties. At CMKC, patient charts were also screened for age of menarche to identify potentially eligible participants and their families attending a scheduled clinical visit. Adolescents were eligible to enroll if they had achieved menarche within 1 year of enrollment, were not taking medications or supplements (ie, hormonal contraceptives, inositol), and did not have any current health condition known or suspected to interfere with reproductive or metabolic health (ie, diabetes, thyroid dysfunction, active eating disorder) or study participation (ie, blood clotting disorders). The potential participant and at least 1 parent/legally authorized representative attended an initial enrollment visit where eligibility was assessed and permission and assent were obtained. Thereafter, a detailed menstrual, medical, and family history was collected in a structured interview. The parent/legally authorized representative completed a demographic survey on behalf of the participant and household. Participants were given a menstrual tracking calendar and instructed to record their menstrual cycles either in the calendar provided or in an electronic app of their preference.

After the initial enrollment visit, up to 4 study visits corresponding to specific gynecological ages were planned. Study visits were set to occur at the following gynecological ages: 1) less than 1 year postmenarche (6–10 months postmenarche; "study visit 1"), 2) gynecological age 1 year (11–13 months postmenarche; "study visit 2"), 3) gynecological age 1.5 years (17–19 months postmenarche; "study visit 3"), and 4) gynecological age 2 years (23–25

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months postmenarche; “study visit 4”). To maximize flexibility for participants and promote retention, study visits for each gynecological age could be scheduled anytime from 1 month before to 1 month after the specified gynecological age. To maximize recruitment and account for COVID-19 pandemic-related shutdowns in research, the study protocol was amended so that participation could begin at a gynecological age of 1 year (corresponding with study visit 2) or 1.5 years (corresponding with study visit 3).

Each study visit involved a virtual pre-study visit and an in-person study visit at either the Cornell Human Metabolic Research Unit or CMKC Pediatric Clinical Research Unit. Participants with regular menstrual cycles were invited to participate within 10 days of the onset of menses, and those with irregular menstrual cycles were invited to participate at any time. In the virtual pre-study visit, participants reported their menstrual cycle diaries and underwent a review of their recent reproductive and medical history, as well as a researcher-guided scoring of their self-reported hirsutism[23] and acne[24] using visual scales. Participants self-reported their Tanner stages using an electronic survey. At the in-person study visit, participants underwent:1) a fasting blood draw, 2) a transabdominal ultrasound scan of the ovaries (GE Voluson E10, RM6C or GE Voluson E8, RAB4-8D), 3) a vital signs assessment, and 4) an anthropometry evaluation, which included use of bioimpedance to measure body composition (Tanita), waist and hip circumference to compute the waist-to-hip ratio (WHR), height, and weight. Study data were collected and managed using REDCap (Research Electronic Data Capture[25]) electronic data capture tools hosted at CMKC.[26,27] An overview of the study design and procedures is provided in Supplementary Table 1.

Ovarian Ultrasonography and Image Analysis

Three-dimensional (3D) transabdominal ultrasound volumes of the ovaries were obtained by a licensed sonographer or trained clinical investigator using a standardized scanning protocol. In brief, sonographers identified the largest cross-sectional view of each ovary and optimized image settings in 2-dimensional (2D) real-time viewing before engaging the 3D modality setting and capturing volumes. Sonographers captured multiple volumes of each ovary and modified standard image settings (eg, gain, depth, focal zone) as needed, with the goal of optimizing resolution of antral follicles and the ovarian perimeter. Volumes were then partitioned into their 2D cineloop components (ie, each of the 3 perpendicular planes) and exported for offline analysis. The 2D cineloop with the best image quality, as judged by the offline rater, was selected for post-hoc image analysis. Images of the ovaries were analyzed by 1 of 3 trained raters. All measurements were double-checked by a senior member of the research team. Grid analysis was used to reliably count the number of antral follicles in each ovary as described elsewhere.[26] Raters completed an internal training course in offline ovarian image analysis, which culminated in an internal reliability study wherein raters assessed FNPO in an independent set of images. Achievement of inter-rater agreement above 0.800 for FNPO was considered the criterion for pooling of data across raters. The primary ultrasonographic endpoint for this study was 2- to 9-mm follicles per ovary (ie, FNPO), and agreement across the 3 raters was 0.814 for this metric. Secondary ultrasonographic endpoints included 1) OV, 2) 2- to 5-mm follicles per ovary ($FNPO_{2–5mm}$), and 3) 6- to 9-mm follicles per ovary ($FNPO_{6–9mm}$), the latter representing 2 physiologically relevant follicle pools: recruitable and selectable follicles, respectively. OV was calculated

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using the equation [π*(average of all 4 linear measurements in orthogonal planes)^3]. For all ultrasonographic endpoints, the mean of both ovaries was computed and recorded for analyses unless otherwise noted. Ultrasound image quality was graded as excellent, partially visible, or poor on the basis of the internally developed criteria provided in Supplementary Table 2.

Biochemical Measurements

Sera were assayed for luteinizing hormone, follicle-stimulating hormone, estradiol, insulin, and sex hormone–binding globulin at the Human Nutritional Chemistry Service Laboratory at Cornell University using chemiluminescent immunoassays (Immulite 2000, Siemens Medical Solutions Diagnostic, Deerfield, IL). The intra- and interassay coefficients of variation (CVs) ranged from 3% to 6% and 5% to 10%, respectively across all analytes. Anti-Müllerian hormone (AMH) was measured using the picoAMH enzyme-linked immunosorbent assay at Ansh Labs (Webster, TX) with intra- and interassay CVs of 2.9% and 5.7%, respectively. Total testosterone was measured via liquid chromatography with tandem mass spectrometry at Brigham and Women's Hospital Research Assay Core (Boston, MA; interassay CV 6.4%), which has been certified by the Centers for Disease Control's Hormone Standardization Program. The free androgen index was computed as [(total testosterone [nmol/L]/sex hormone–binding globulin [nmol/L]) × 100], and bioavailable and free androgens were determined using calculations described previously.[27] Fasting glucose was measured using a standard glucometer (Accu-Chek Aviva, Roche, Basel, Switzerland), and lipids were measured using a DIMENSION XPAND Plus Chemistry Analyzer (Siemens Medical Solutions Diagnostic) with intra- and interassay CVs of less than 4% and less than 1.5%, respectively.

Cross-sectional Analysis of Study Visit 1 Measures (6–10 months postmenarche)

Data from participants who completed study visit 1, for which sera and sufficient menstrual cycle data were available, were included as part of this pilot analysis (N = 31). An inter-menstrual interval was defined as the first day of menses to the day preceding the subsequent menses. Menses was defined as at least 2 sequential days of self-reported bleeding. The average menstrual cycle length was established using up to 3 inter-menstrual intervals obtained from prospectively maintained menstrual diaries leading up to study visit 1.

Statistical Analyses

Spearman rank correlation coefficients were used to assess bivariate associations between ovarian endpoints or menstrual cycle length with metabolic and reproductive endpoints. Linear regression analyses were conducted to test the independent predictors of ovarian morphology using variable selection from bivariate correlations of a P value less than 0.1.

Ethical Considerations

The study was approved by the CMKC Institutional Review Board (IRB#00000779) with reliance from the Cornell University Institutional Review Board. Permission was obtained from one parent/legally authorized representative and assent from each adolescent participant before study procedures were conducted.

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Results

Forty participants enrolled in the OMG! study, with 33, 4, and 3 participants initiating the study at 6–10 months postmenarche, 11–13 months postmenarche, and 17–19 months postmenarche, respectively (Fig. 1). To balance study feasibility amid the COVID-19 pandemic (ie, repeated clinical research shutdowns, study timeline, and depletion of funds), the study team ended enrollment at 40 participants. Of the 33 participants enrolled at study visit 1, 2 participants were excluded from the current analysis because a blood sample could not be obtained (n = 1) or, at the time of sera assay, the participant had not yet completed study visit 1 (n = 1).

Participant characteristics are summarized in Table 1. As a group, participants were approximately 13 years old, identified as White (25/31, 81%), non-Hispanic (29/31, 94%), and had high socioeconomic status (household income >$100,000/year; 20/31, 64%). Participants were primarily lean in body composition and had glucoregulatory status markers and lipid levels on average within the normative ranges.

On transabdominal ultrasonography, the left and right ovaries were judged as being partially visible or having excellent visibility in 90% and 78% of participants, respectively (Fig. 2). Measurements of FNPO and OV were possible in 30 of 31 participants, as summarized in Supplementary Table 3.

Both OV and FNPO were positively associated with AMH and negatively associated with fasting insulin. A positive trend was detected between FNPO and total testosterone and a negative trend detected with WHR (Table 2). Linear regression models confirmed that AMH and WHR were significant independent predictors of FNPO, whereas AMH and fasting insulin significantly predicted OV (Table 3). The average menstrual cycle length was only negatively associated with triglycerides levels (Tables 2 and 3).

Discussion

The present study aimed to establish whether ovarian morphology on ultrasonography reflected reproductive and metabolic features within the first postmenarcheal year. In partial support of our hypothesis, we found that ovarian features were readily resolved in most participants, with antral follicle numbers and ovarian size positively related to AMH and inversely related to insulin. This was in contrast to average menstrual cycle status, which did not reflect any reproductive parameter. Together, these data point to the potential for ovarian morphology as an early, noninvasive biomarker of reproductive function.

Our findings align with previous work in perimenarcheal adolescents showing that ovarian morphology reflects serum AMH concentrations.[28,29] Among ultrasonographically detectable follicles, AMH is produced by 2- to 9-mm follicles,[30] which were readily captured in the present study. This study demonstrates that modern transabdominal ultrasound enables detection of well-established relationships between endocrinology and morphology (ie, AMH and FNPO) and extends previous findings[31] to demonstrate that AMH can reflect follicle number and OV during the first gynecological year. By contrast, FNPO only tended to correlate with total testosterone, but there were no associations

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between FNPO or OV and free androgens, which partially reflects our previous reports of positive associations of FNPO and ovarian size with androgens in adolescents with PCOS.[11] Although it is possible that the failure to detect previously reported associations may be a type II error attributed to a small sample size, we suspect that we simply did not capture a sufficient spectrum of androgen levels in our cohort; the adolescents in the present study did not exhibit overt evidence of hyperandrogenism despite a wide range in FNPO and OV. Previous research suggests that androgens increase during the postmenarcheal years.[32] Therefore, it is plausible that our assessments were timed well before any progression to ovarian hyperandrogenism.

Consistent with our previous studies in adults[14] and adolescents[11] with PCOS, follicle populations and ovarian size were associated with markers of metabolic status. Namely, we report similar inverse relationships between insulin, central adiposity, and ovarian morphology even as early as 6–10 months postmenarche. We have hypothesized that there are both suppressive and stimulatory effects of impaired glucoregulation and obesity on the reproductive axis.[15] Although it is possible that the predictive nature of WHR for FNPO could be attributed to impaired resolution of small antral follicles with obesity, similar associations have been reported using transvaginal ultrasonography in adults.[14,32] Failure to detect an association between fasting glucose and lipids with ovarian features may be explained by insulin having a more defined direct effect[33] on the reproductive axis, whereas dysglycemia and dyslipidemia may reflect broader metabolic complications that indirectly impact ovarian function. Together, we interpret our association between ovarian morphology and markers of metabolic status to be suggestive of a physiological linking of metabolic health to reproductive development and not a technical artifact.

The International Guideline for Diagnosis of PCOS does not advise ultrasound for the diagnosis of PCOS in adolescents owing to the reduced image quality of transabdominal ultrasonography and polycystic-like ovarian morphology during the early postmenarcheal years.[34] This study supports a growing consensus[12,33] that modern transabdominal ultrasonography can adequately resolve antral follicles, thereby providing a more comprehensive evaluation of ovarian features.[13,29] The subset of ovaries deemed to have poor image quality was attributed to anatomical or physiologic limitations (ie, active bowels, constipation) and not reflective of imaging capabilities per se. Indeed, use of higher frequency transabdominal probes, such as 3–8 MHz, is appropriate across a range of body sizes.[28,29] In light of our current findings, we propose that consideration of ovarian morphology in the evaluation of PCOS in adolescence is no longer a constraint of transabdominal ultrasound image quality. Rather, normative data in this age group are needed to define the upper limits of physiologic changes in ovarian morphology during adolescence.[13,20,28,35] To that end, we strongly discourage the application of adult criteria for polycystic ovarian morphology in adolescents and instead advocate for prospective data collection to develop developmental stage–specific criteria.

This study represents the analysis of data from the first and only prospective cohort study of extensively phenotyped US-based adolescents beginning in the first gynecological year. These data provide promising insight into the utility of ovarian morphology as a biomarker for early reproductive dysfunction. This study also has limitations. The cohort is not

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representative of the US demographic makeup given that most participants identified as non-Hispanic White from a household with high socioeconomic status. Unintentionally, this study also captured the adolescent reproductive transition during a global pandemic, which acutely altered the environment and routines of adolescents. Because of the pandemic, we modified our original study protocol to limit in-person interactions, which included a transition to virtual clinical assessments (ie, hirsutism, acne, Tanner staging) using researcher-guided self-reporting. Although this hybrid approach was adopted across both sites to prioritize data consistency and enable feasibility, self-reporting may be prone to increased variability in certain measures. Likewise, depletion of funds did not enable us to reach our intended sample size. Although our a priori sample size calculation was made on the basis of limited data in non-domestic populations, we acknowledge that our study sample may have been insufficient to capture differences between groups. Nonetheless, we view our exploratory analysis as an important first step to establish larger studies, of sufficient duration, to characterize the functional and morphologic trajectories experienced by our current adolescent population.

In summary, ovarian morphology reflected reproductive and metabolic status markers, specifically AMH and fasting insulin, in the first postmenarcheal year. Future reports involving the OMG! cohort will define the changes in ovarian morphology that accompany the establishment of regular vs irregular cycles and early evidence of PCOS risk. Additional research is needed to establish the natural trajectory of reproductive maturation in a modern, diverse cohort to best inform early, robust biomarkers of PCOS risk.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

We are grateful to the staff of the Pediatric Clinical Research Unit (CMKC) and the Human Metabolic Research Unit (Cornell University) for use of the facilities and staff support. We are also grateful to the Department of Pediatric Endocrinology and Adolescent Medicine (CMKC) for their willingness to support recruitment and the success of OMG!. We also acknowledge and express gratitude to Dara Watkins, the research coordinator at Children's Mercy, who supported much of the data collection in Kansas City. We also acknowledge the graduate students and undergraduate students, and the Cornell Statistical Consulting Unit, who supported different aspects of the study design, analysis, and data management. We are grateful to Brittany Jarrett, who assisted with the development of the image quality matrix. Finally, we are extremely grateful to the incredible families and participants who enrolled in OMG!, despite the challenges of a global pandemic; this research would not have been possible without them. Data were presented as a poster at the Endocrine Society Annual Meeting (June 10–14, 2022, Atlanta, GA; Abstract SUN-373) and as an oral presentation at the Androgen Excess and PCOS Society Annual Meeting (October 21–23, 2022, Anaheim, CA).

Funding Statement

Funding was provided by the National Institutes of Health (R21-HD095372), Canadian Institutes of Health Research (Postdoctoral Fellowship #171268; Vanden Brink) 1K23HD109464-01, and National Institutes of Child Health and Human Development (Barral). The study sponsors had no role in the design, data collection or analysis, writing, or decision to submit for publication.

Data Availability

Data will be made available to the editors of the journal for review or query upon request.

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Abbreviations:

AMH	Anti-Mullerian Hormone
BMI	Body Mass Index
FNPO	Follicle Number Per Ovary
FSH	Follicle Stimulating Hormone
HDL	High Density Lipoprotein
LDL	Low Density Lipoprotein
LH	Luteinizing Hormone
MCL	Menstrual Cycle Length
OV	Ovarian Volume
SHBG	Sex Hormone Binding Globulin
T	Testosterone

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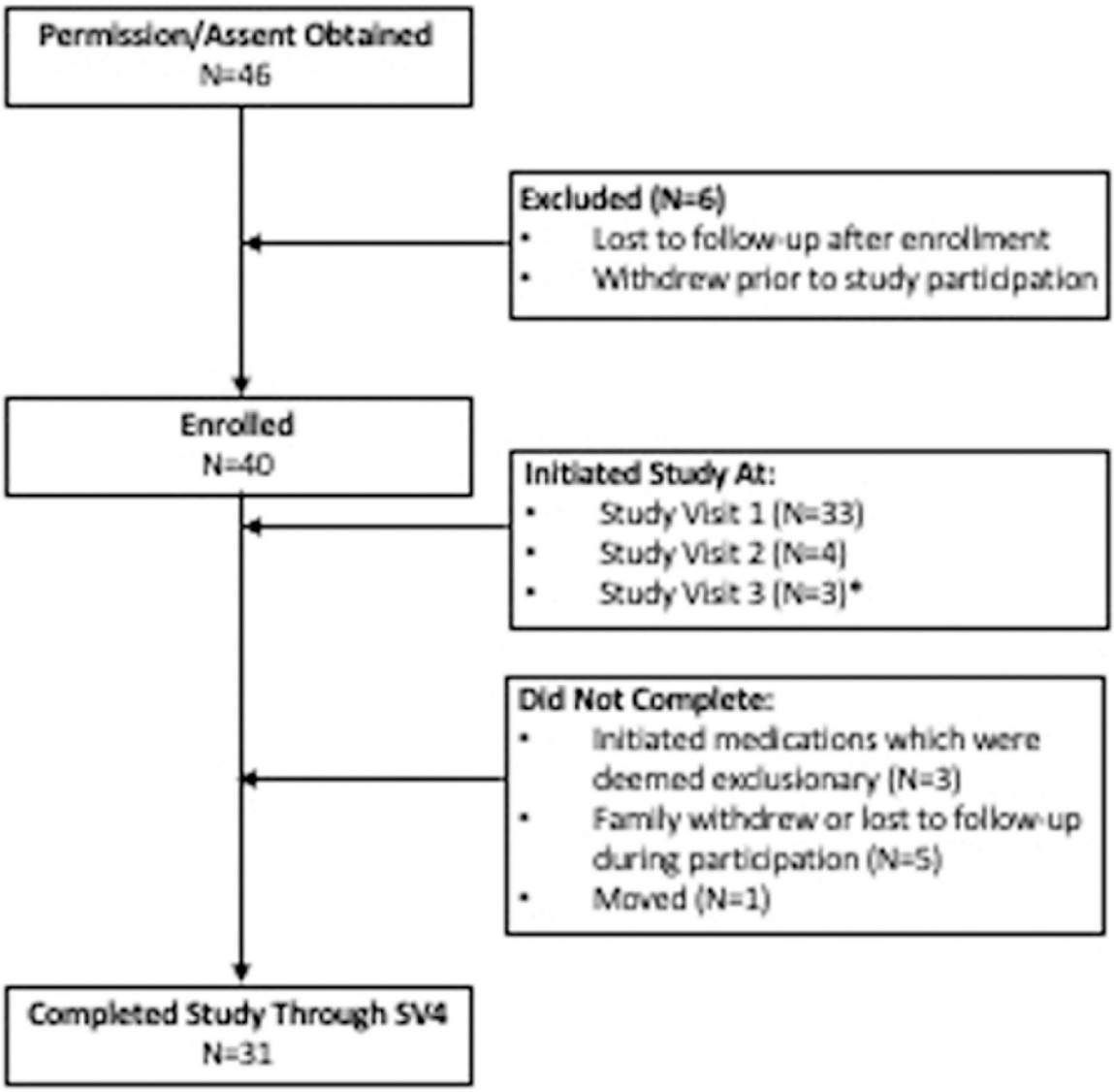

Fig. 1.

Participant flow chart of those enrolled at Children's Mercy Kansas City and Cornell University. *Due to the COVID-19 global pandemic and clinical research shutdowns at both sites, participants who were already enrolled were permitted to begin the study at the next gynecological timepoint once research was permitted to resume. Thirty-two participants were enrolled in Kansas City, MO, and 8 were enrolled in Ithaca, NY.

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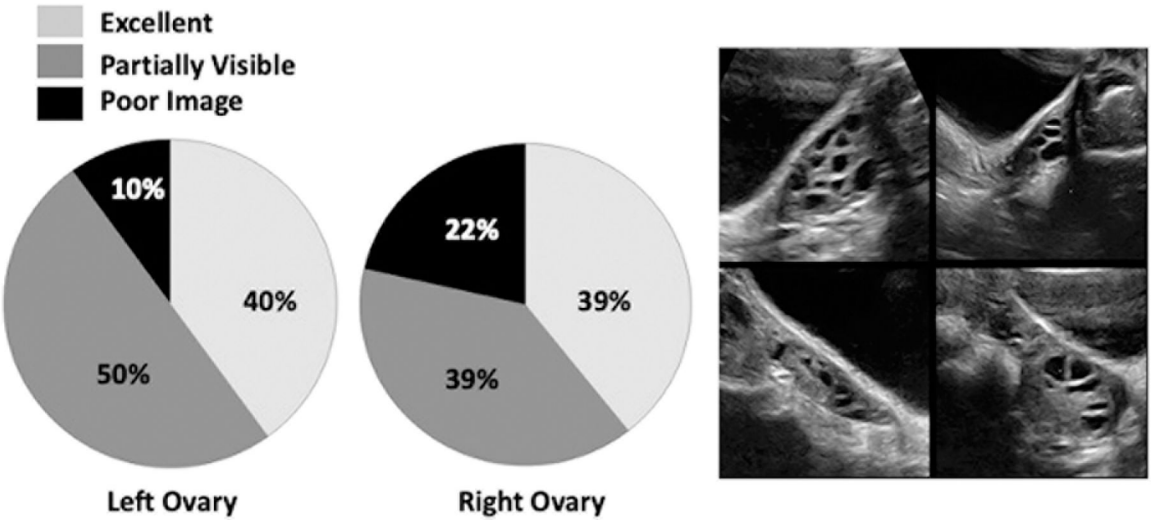

Fig. 2.

Image quality of ovaries obtained via transabdominal imaging. Image quality of the left and right ovaries analyzed was evaluated using an internal rubric (Supplementary Table 1). Examples of excellent image quality obtained using transabdominal ultrasound imaging are shown on the right.

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Table 1

Characteristics of the Participants at Study Visit 1 Corresponding to 6–10 Months Postmenarche (N = 31).

	Median	Interquartile range
Chronological age (months)	158.7	(149.0–164.5)
Gynecological age (months)	6.9	(5.8–8.0)
Race (N, %)		
White		25/31 (81)
Asian		3/31 (10)
Black or African American		0/31 (0)
American Indian and Alaska Native		0/31 (0)
Hawaiian and other Pacific Islander		0/31 (0)
More than one race		1/31 (3)
Unknown or not reported		2/31 (7)
Ethnicity (N, %)		
Hispanic		1/31 (3)
Not Hispanic		29/31 (94)
Not reported		1/31 (3)
Household income (N, %)		
<$20,000/y		0/31 (0)
$20,000–$39,999/y		0/31 (0)
$40,000–$59,999/y		0/31 (0)
$60,000–$79,999/y		4/31 (13)
$80,000–$99,999/y		4/31 (13)
>$100,000/y		20/31 (64)
Not reported		3/31 (10)
Anthropometry		
Body mass index (kg/m^2)	20.2	(18.3–22.9)
Waist-to-hip ratio	0.7	(0.7–0.8)
Total body fat (%)	24.9	(21.9–30.4)
Metabolic features		
Fasting glucose (mg/dL)	94	(89.5–106)

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	Median	Interquartile range
Fasting insulin (uIU/mL)	7.8	(6.4–11.4)
SHBG (nmol/L)	53.7	(43.2–62.0)
HDL (mg/dL)	54.55	(48.6–57.4)
LDL (mg/dL)	85.2	(81.1–101.4)
Cholesterol (mg/dL)	158.4	(146.6–171.5)
Triglycerides (mg/dL)	68.4	(48.8–92.1)
Reproductive features		
Average cycle length (d)	34	(26–55)
Acne (N, %)		
No acne observed		4/31
Grade 1		12/31
Grade 2		4/31
Grade 3		10/31
Grade 4		1/31
Hirsutism score	1	(0–3)
LH (mIU/mL)	3.5	(3.0–4.8)
FSH (mIU/mL)	5.2	(4.4–6.2)
Estradiol (pg/dL)	39.8	(34.4–44.3)
Free testosterone (ng/dL)	0.32	(0.23–0.45)
Bioavailable testosterone (ng/dL)	7.5	(5.3–10.5)
Total testosterone (ng/dL)	24.7	(22.1–34.8)
Ovarian features		
AMH (ng/mL)	4.5	(2.4–6.0)
OV (cm^3)	6.8	(4.5–8.3)
FNPO 2–9 mm	23	(12–27)
FNPO 2–5 mm	16	(9–20)
FNPO 6–9 mm	4	(2–9)

AMH, anti-Müllerian hormone; FNPO, follicle number per ovary; FSH, follicle-stimulating hormone; HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol; LH, luteinizing hormone; OV, ovarian volume; SHBG, sex hormone–binding globulin.

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Table 2

Bivariate associations between menstrual cycle length or ovarian morphology with reproductive and metabolic features.

	Average Cycle Length	OV	FNPO
Chronological Age (months)	0.141	0.242	0.275
Gynecological Age (months)	−0.145	−0.193	−0.279
Metabolic Features			
BMI (kg/m^2)	−0.041	−0.071	−0.113
Fasting Glucose (nmol/L)	0.151	0.029	0.078
Fasting Insulin (uIU/mL)	−0.169	**−0.514** **	**−0.419** *
Waist-to-Hips Ratio	−0.287	−0.270	−0.322^t
Total Body Fat (%)	−0.223	−0.195	−0.189
SHBG (nmol/L)	0.049	−0.127	−0.170
HDL (mg/dL)	0.116	−0.124	0.048
LDL (mg/dL)	−0.078	−0.049	0.073
Cholesterol (mg/dL)	−0.054	−0.228	0.098
Triglycerides (mg/dL)	**−0.386** *	−0.010	0.196
Reproductive Features			
Hirsutism Score	0.102	0.197	0.048
FSH (mIU/mL)	−0.021	−0.251	−0.269
LH (mIU/mL)	0.232	0.116	0.066
Total Testosterone (ng/dL)	0.022	0.182	0.339^t
Free Testosterone	0.170	0.031	0.179
Bioavailable Testosterone	0.177	0.027	0.185
AMH (ng/mL)	0.124	**0.551** ***	**0.668** ***

Spearman's Rank Correlation Coefficients:

^{t}P<0.10

*P<0.05

**P<0.01

***P<0.001

Abbreviations: AMH, Anti-Mullerian Hormone; BMI: Body Mass Index; FNPO, Follicle Number Per Ovary; FSH, Follicle Stimulating Hormone; HDL, High Density Lipoprotein; LDL, Low Density Lipoprotein, LH, Luteinizing Hormone; OV, Ovarian Volume; SHBG, Sex Hormone Binding Globulin; T, Testosterone.

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Table 3

Independent Predictors of Ovarian Morphology in the First Postmenarcheal Year*.

	Predictors	Estimate	*P* value
Follicle number per ovary	**AMH**	0.520	**.001**
	Total testosterone	0.181	.309
	Waist-to-hip ratio	−3.223	**.015**
	Fasting insulin	0.008	.964
Ovarian volume	Average cycle length	−0.001	−.761
	AMH	0.332	**.006**
	Fasting insulin	−0.323	**.039**

AMH, anti-Müllerian hormone.

*In all models, all variables except for average cycle length were log transformed to meet model assumptions.